

		kinetic energy of the objects, hence kinetic energy is not conserved.
7	D	Using conservation of energy, $mg\Delta h = \frac{1}{2}mv^2$ $v^2 = 2g\Delta h$ $u = \sqrt{2 \times 9.81 \times 3.0} = 7.672 \text{ m s}^{-1} \text{ (downward)}$ $v = \sqrt{2 \times 9.81 \times 2.0} = 6.264 \text{ m s}^{-1} \text{ (upward)}$ By Newton's second law, average resultant force on the ball when in contact with the ground $\langle F \rangle = \frac{\Delta p}{\Delta t} = \frac{(0.050)[6.264 - (-7.672)]}{0.010}$ $= 69.68$ $\approx 70 \text{ N}$
8	B	